

ALEX SHONENKOV

Senior Applied Research Engineer, 10+ years of experience
GMT+4

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[Technical Portfolio](#)

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ABOUT

Senior Applied Research Engineer with 10+ years of experience in machine learning, large-scale ML systems, generative models, and production-grade research pipelines. Experienced in turning research ideas into reliable training, evaluation, inference, and release workflows for open-source releases and production ML systems.

Strong engineering foundation in Python, PyTorch, distributed training, data pipelines, and ML infrastructure, with systems-level experience in Rust and Maturin for optimizing Python-based research workflows.

Kaggle Code Grandmaster with practical experience across computer vision, NLP, time series, multimodal learning, and competitive machine learning.

SKILLS

Core: Generative AI, Multimodal Systems, Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision (CV), Time Series, Applied Math, Research and Development (R&D), Data Mining, Large-Scale Distributed Training

Tech: Python, PyTorch, Rust, Maturin, NumPy, SciPy, scikit-learn, CatBoost, LightGBM, XGBoost, Hugging Face, Transformers, Diffusers, Jupyter, WandB

Data: Pandas, Polars, PySpark, ClickHouse, SQL, DuckDB, WebDataset, AWS S3

Others: Docker, Linux, Git, CI/CD, Torch-XLA, DeepSpeed, Megatron-LM, Grafana, Business Intelligence (BI), REST API

EXPERIENCE

Independent Applied Research — Quantitative ML Systems

04/2024 – Present

Senior Applied Research Engineer

- Built a quantitative ML research stack for crypto markets, processing multi-terabyte raw trade data from 2011 onward across 8 markets and 60+ coins.
- Developed real-time and historical data pipelines for raw trade ingestion, OHLCV candle generation, and second-level VWAP-based index price calculation across markets.
- Built a backtesting and trading terminal supporting live Bybit trading, leakage-safe historical replay with time controls, AI strategy plugins, and headless metric evaluation.
- Developed performance-critical Rust components for market indicators, backtesting, and ClickHouse UDFs, enabling fast SQL-based trading metric analysis in Grafana.

Technologies: Python, Rust, Maturin, ClickHouse, DuckDB, Polars, PyTorch, Time Series, backtesting, feature engineering, trading systems, Data Mining

Stability AI

04/2022 – 04/2024, 2 years

Senior Applied Research Engineer

- Developed core components of DeepFloyd IF, a 4.3B-parameter text-to-image diffusion model with cascaded super-resolution stages, released as a major open-source generative AI model with 7.8k+ GitHub stars.
- Built large-scale distributed training and evaluation pipelines on 512 A100 GPUs for pixel-space diffusion models conditioned on T5 text embeddings.
- Contributed to Hugging Face Diffusers integration, public demos, and reproducible examples, improving accessibility and adoption of the open-source model release.
- Developed features for internal Stable Diffusion training infrastructure, improving experimentation workflows for large-scale text-to-image model training.

Technologies: Python, PyTorch, WandB, WebDataset, AWS S3, Transformers, Generative AI, Multimodal Systems, mixed precision, Docker, DeepSpeed, Git

Sber AI

06/2021 – 04/2022, 1 year

Senior Applied Research Engineer

- Developed core components of ruDALL-E, among the first Russian-language DALL-E-style text-to-image model families, helping establish an open-source foundation for Russian multimodal generation with 1.6k+ GitHub stars.
- Built large-scale distributed training, evaluation, and inference pipelines on 512 V100 GPUs for architectures based on DALL-E, CLIP, and VQ-GAN, turning research prototypes into reproducible training and release workflows.

- Developed memory-optimization techniques for large-scale neural network training, including Few-bit Backward, a novel activation-gradient quantization method published at ICML 2023.
- Contributed to ruCLIP, RUDOLPH, and Tuned VQ-GAN open-source releases, expanding the Russian multimodal stack across image-text retrieval, generation, captioning, and visual tokenization.
- Released Digital Peter and StackMix-OCR open-source projects for historical handwritten text recognition, including competition baselines, training pipelines, and reproducible notebooks, with related HIP@ICDAR 2021, arXiv, and Computer Optics publications.

Technologies: Python, PyTorch, Transformers, Megatron-LM, Hugging Face, PyTorch Lightning, DeepSpeed, DDP, GPT, WebDataset, PySpark, AWS S3, Generative AI, Multimodal Systems, Docker, CI/CD

OCRV (Russian Railways Affiliated Company)

01/2019 – 06/2021, 2.5 years

Senior ML / Data Engineer

- Achieved Kaggle Code Grandmaster tier and ranked #5 worldwide during this period.
- Mentored AI students at the Sirius Educational Center, guiding ML research and engineering projects.
- Developed production computer vision systems for inspection of railway infrastructure.
- Designed multimodal ML pipelines processing images, text, video and tabular data.
- Delivered machine learning models from research prototypes to production deployment.

Technologies: Python, PyTorch, scikit-learn, Transformers, Deep Learning, Flask, SQL, GitLab CI/CD, Docker, Grafana, OCR, object detection, text classification, text embedders, Research and Development (R&D), LightGBM, XGBoost, Torch-XLA, Business Intelligence (BI)

Lia Chat

05/2017 – 01/2019, 1.5 years

Data / ML Engineer

- Developed NLP models for text classification, clustering, named entity recognition (NER), relevance ranking, and embedding-based retrieval.
- Built machine learning microservices for ETL, model inference, model serving, and Telegram bot integrations.
- Designed production pipelines for deploying and serving NLP models in AI-powered virtual assistant applications.
- Contributed to a scalable microservices architecture and production machine learning systems.
- Worked in a small, highly technical team delivering production AI solutions for business customers.

Technologies: Python, PyTorch, TensorFlow, Jupyter, Flask, Django, asyncio, aiohttp, PostgreSQL, MongoDB, GitLab CI/CD, Docker, SQL, Nginx, Applied Math, Data Mining, NumPy, SciPy, scikit-learn, CatBoost, Linux, Git, REST API

Joint Institute for High Temperatures

10/2016 – 07/2018, 2 years

Researcher (Part-time position)

Conducted research at the Joint Institute for High Temperatures of the Russian Academy of Sciences (JIHT RAS), investigating the influence of dust particles on neon spectral line intensities in the positive column of a DC discharge under microgravity conditions. Developed ETL pipelines and data processing scripts for the Plasma Kristall-4 space experiment, processing approximately 1 TB of text logs and images, and performed exploratory data analysis (EDA). Developed expertise in large-scale data processing, pattern discovery in multimodal datasets, and scientific writing using LaTeX.

MIPT Business Incubator

06/2016 – 05/2017, 1 year

Junior Data Analyst

At the Moscow Institute of Physics and Technology (MIPT), I gained my first professional experience working with data. My responsibilities included analyzing startup technologies, conducting market research, collecting and visualizing data. Developed strong research skills, analytical thinking, and a solid foundation in programming.

Institute of Spectroscopy

09/2014 – 06/2016, 2 years

Junior Researcher (Part-time position)

Conducted research at the Institute of Spectroscopy of the Russian Academy of Sciences, focusing on the emission spectroscopy of H₂ molecules in neon lamps. Contributed to the development of a universal calibration system for emission spectroscopy instruments, making calibration more accessible and easier to use.

Gained hands-on experience in rapid prototyping and scientific research.

EDUCATION

Moscow Institute of Physics and Technology (State University) (MIPT)
Master of Science (MSc) 03.04.01 Applied Mathematics and Physics, 2012–2018

ADDITIONAL INFORMATION

Outside of professional work, I enjoy chess, snowboarding, playing the piano, and homelab GPU computing.